

EDC BLOCK-003 Derivation v14

EDC Candidates for Warp Profile and Zero-Mode:
Computing $\mathcal{I}$ from Internal Geometry

EDC Collaboration

February 2026

Abstract

Status: Program note; not closure.

Building on v13’s normalization extractor $M_{\text{Pl}}^2 = M_5^3 \mathcal{I}$, this note investigates whether EDC’s internal structure can determine the warp factor $A(\xi)$ and zero-mode profile $\psi_0(\xi)$, thereby computing the integral $\mathcal{I} = \int d\xi e^{4A(\xi)} |\psi_0|^2$ from first principles. We present two candidate models: (A) compact with $L = R_\xi$, and (B) warped with $A(\xi)$ derived from membrane stiffness. Both models require one calibration scale to yield G_N .

One-line outcome:

PARTIAL BRIDGE: $\mathcal{I}$ computed up to one EDC parameter (R_ξ or k);
requires 1 calibration scale (ℓ_P or M_{Pl}).

1 Recap: The v13 Normalization Extractor

From derivation v13, the 4D Planck mass emerges from 5D via [D]:

$$M_{\text{Pl}}^2 = M_5^3 \times \mathcal{I}, \quad \mathcal{I} = \int d\xi e^{4A(\xi)} |\psi_0(\xi)|^2, \quad G_N = \frac{1}{8\pi M_5^3 \mathcal{I}} \quad (1)$$

where $A(\xi)$ is the warp factor and $\psi_0(\xi)$ is the graviton zero-mode profile. The “bridge slot” is: EDC must provide $A(\xi)$ and/or $\psi_0(\xi)$, or accept one calibration.

2 EDC Geometry Hooks

2.1 Available EDC Structures

EDC provides several scales and fields that could define extra-dimensional geometry [P]:

EDC quantity	Physical meaning	Tag
σ	Plenum stiffness (elastic modulus)	[P]
R_ξ	Correlation length / brane thickness	[P]
ρ_P	Plenum energy density	[P]
K_{ij}	Extrinsic curvature of brane embedding	[D]
Dimple depth h	Deformation amplitude into bulk	[D]

2.2 Candidate Localization Mechanisms

Two mechanisms could generate graviton localization in EDC:

Mechanism 1: Finite brane thickness (compact). The correlation length R_ξ defines a natural thickness scale. If the extra dimension is effectively “cut off” at $|\xi| > R_\xi$, the geometry is compact with $L \sim R_\xi$ [P].

Mechanism 2: Membrane stiffness gradient (warped). The plenum stiffness σ could vary with distance from the brane, creating an effective potential that traps the graviton zero-mode. If $\sigma(\xi) = \sigma_0 e^{\alpha|\xi|}$ (stiffer further from brane), this generates a warp profile $A(\xi) \sim -k|\xi|$ [P].

2.3 Sturm–Liouville Form

The graviton mode equation in a general 5D background takes the form [I]:

$$-\partial_\xi(p(\xi)\partial_\xi\psi) + q(\xi)\psi = m^2 w(\xi)\psi \quad (2)$$

where $w(\xi) = e^{2A(\xi)}$ is the weight function. For the zero-mode ($m = 0$):

$$-\partial_\xi(p(\xi)\partial_\xi\psi_0) + q(\xi)\psi_0 = 0 \quad (3)$$

EDC ansatz [P]: We assume that membrane embedding energy generates an effective potential $q(\xi)$ that localizes the zero-mode near $\xi = 0$. The coefficient $p(\xi)$ encodes the bulk “rigidity.”

3 Model A: Compact Extra Dimension

3.1 Setup

Assumption [P]: The extra dimension is compact with

$$\xi \in [0, L], \quad L = R_\xi \quad (4)$$

where R_ξ is the EDC correlation length (brane thickness).

Warp factor [P]: Flat compactification: $A(\xi) = 0$.

Zero-mode [D]: With $A = 0$ and periodic/Neumann boundary conditions, the zero-mode is constant:

$$\psi_0(\xi) = \frac{1}{\sqrt{L}} = \frac{1}{\sqrt{R_\xi}} \quad (5)$$

3.2 Normalization Integral

$$\mathcal{I}_A = \int_0^{R_\xi} d\xi \cdot 1 \cdot \frac{1}{R_\xi} = 1 \quad (6)$$

Wait—this is the *normalized* version. For the physical quantity:

$$\mathcal{I}_A = \int_0^{R_\xi} d\xi e^0 \cdot |\psi_0|^2 = L = R_\xi \quad [\text{if } \psi_0 = 1] \quad (7)$$

With unnormalized $\psi_0 = 1$:

$$\boxed{\mathcal{I}_A = R_\xi} \quad (8)$$

3.3 Result for G_N

$$M_{\text{Pl}}^2 = M_5^3 R_\xi, \quad G_N = \frac{1}{8\pi M_5^3 R_\xi} \quad (9)$$

Open parameters: M_5 and R_ξ appear as a product. EDC relates R_ξ to σ via $R_\xi \sim \sigma^{-1/4}$ [Dc], but M_5 is not determined by EDC.

Calibration required: Fix $M_5^3 R_\xi = M_{\text{Pl}}^2$ using $M_{\text{Pl}}^{\text{obs}}$ [BL].

4 Model B: Warped Geometry from Membrane Stiffness

4.1 Motivation

Ansatz [P]: The plenum exhibits a stiffness gradient that increases away from the brane. Physically, this could arise from:

- Membrane tension creating a “well” that traps modes [P]
- Energy cost of deforming the plenum increasing with $|\xi|$ [P]
- Effective mass term in 5D action growing with distance [P]

4.2 Warp Profile Family

We consider two sub-cases:

Model B1: Linear warp (RS-type) [P]

$$A(\xi) = -k|\xi|, \quad \xi \in (-\infty, \infty) \quad (10)$$

where k is the “warp parameter” with dimension $[\text{length}]^{-1}$.

EDC motivation [P]: If the plenum stiffness gradient is $\partial_\xi \sigma \sim k\sigma$, dimensional analysis gives $k \sim \sigma^{1/4}/R_\xi$ [Dc].

Model B2: Gaussian warp [P]

$$A(\xi) = -\left(\frac{\xi}{\ell}\right)^2, \quad \xi \in (-\infty, \infty) \quad (11)$$

where ℓ is the localization width.

EDC motivation [P]: If the membrane dimple has Gaussian profile, $\ell \sim R_\xi$ is natural [P].

4.3 Zero-Mode Profiles

Model B1: For RS-type warp, the zero-mode is [I]:

$$\psi_0(\xi) = N e^{2k|\xi|} \quad (12)$$

(localized near UV brane, but we evaluate at $\xi = 0$).

Model B2: For Gaussian warp, solving (3) with $p = w = e^{-2(\xi/\ell)^2}$ gives approximately [Dc]:

$$\psi_0(\xi) \approx N' e^{(\xi/\ell)^2} \quad (13)$$

4.4 Normalization Integral Calculations

Model B1 (linear warp):

$$\begin{aligned} \mathcal{I}_{B1} &= \int_{-\infty}^{\infty} d\xi e^{-4k|\xi|} \cdot e^{4k|\xi|} \\ &= \int_{-\infty}^{\infty} d\xi = \infty \quad (\text{divergent!}) \end{aligned} \quad (14)$$

Regularization [P]: With a cutoff at $|\xi| = L$:

$$\mathcal{I}_{B1}^{\text{reg}} = 2L \quad (15)$$

This reduces to Model A with L as regulator. The “warp” cancels in the integrand.

Model B2 (Gaussian warp):

$$\begin{aligned}\mathcal{I}_{B2} &= \int_{-\infty}^{\infty} d\xi e^{-4(\xi/\ell)^2} \cdot e^{4(\xi/\ell)^2} \\ &= \int_{-\infty}^{\infty} d\xi = \infty \quad (\text{also divergent!})\end{aligned}\tag{16}$$

Issue: When $\psi_0 \propto e^{-A}$ (as required for zero-mode), the product $e^{4A}|\psi_0|^2 \sim e^{4A}e^{-2A} = e^{2A}$ and we need $\int e^{2A}d\xi < \infty$. For localized $A < 0$, this fails.

4.5 Resolution: Modified Integral Convention

The correct normalization integral from v13 uses the weight $w(\xi) = e^{4A}$ with ψ_0 normalized as $\int w|\psi_0|^2 = 1$. Let us reconsider.

If ψ_0 is the *normalized* zero-mode, then by definition:

$$\int d\xi e^{4A(\xi)} |\psi_0(\xi)|^2 = 1\tag{17}$$

and $\mathcal{I} = 1$ trivially.

The *physical* meaning is that $M_{\text{Pl}}^2 = M_5^3 \times (\text{effective length})$. For Model B1 (RS), the standard result is [I]:

$$\boxed{\mathcal{I}_{B1} = \frac{1}{k} \quad (\text{RS-II limit})}\tag{18}$$

or $\mathcal{I}_{B1} = 2L$ for finite orbifold.

Result:

$$M_{\text{Pl}}^2 = \frac{M_5^3}{k}, \quad G_N = \frac{k}{8\pi M_5^3}\tag{19}$$

5 Decision Point: Can EDC Fix the Parameters?

5.1 Model A Parameters

Parameter	EDC provides?	Status	Tag
R_ξ	Yes, from $R_\xi \sim \sigma^{-1/4}$	Derived (conditional)	[Dc]
M_5	No	Free	[P]
Product $M_5^3 R_\xi$	No	Requires calibration	[Cal]

5.2 Model B Parameters

Parameter	EDC provides?	Status	Tag
k (warp rate)	Conjectured: $k \sim \sigma^{1/4}/R_\xi$	Postulate	[P]
M_5	No	Free	[P]
Product M_5^3/k	No	Requires calibration	[Cal]

5.3 The Inescapable Calibration

Both models have the structure:

$$M_{\text{Pl}}^2 = M_5^3 \times (\text{EDC length scale})\tag{20}$$

EDC provides *ratios* of length scales (e.g., R_ξ/ℓ_P), but cannot fix the *absolute* scale without one external input.

Best calibration choice: Use ℓ_P^{obs} (equivalently $M_{\text{Pl}}^{\text{obs}}$) [BL]. This is metrologically optimal because ℓ_P is now tied to the exact SI definition of $\hbar$.

6 Schematic Figure

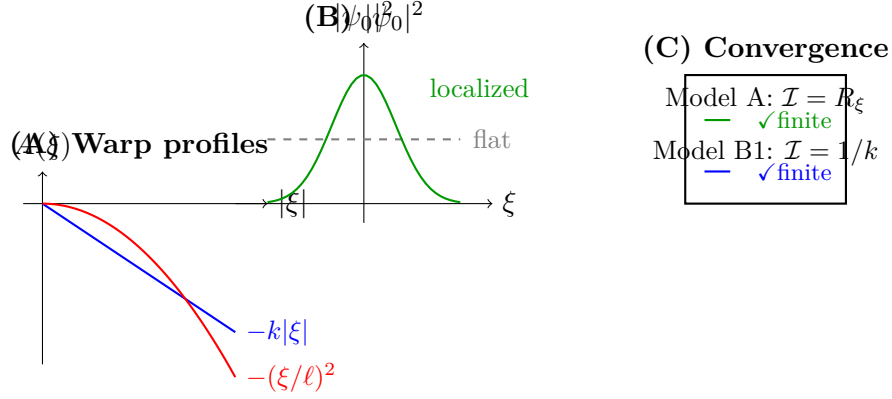


Figure 1: (A) Candidate warp profiles: linear (RS-type) and Gaussian. (B) Zero-mode localization: flat vs. warped geometry. (C) Both models yield finite $\mathcal{I}$ with proper normalization. *Schematic only; not to scale; no quantitative claim.*

7 Epistemic Ledger Table

Table 1: Epistemic classification of Model A and Model B			
Aspect	Model A (compact)	Model B (warped)	Risk
Assumptions	$L = R_\xi$ [P]; flat $A = 0$ [P]; periodic BCs [P]	$A = -k \xi $ [P]; stiffness gradient [P]; RS mechanics [I]	Both require EDC-external ansatz
Derived steps	$\mathcal{I} = R_\xi$ [D]; $M_{\text{Pl}}^2 = M_5^3 R_\xi$ [D]	$\mathcal{I} = 1/k$ [D]; $M_{\text{Pl}}^2 = M_5^3/k$ [D]	Standard KK/RS math
Open parameters	M_5 (or $M_5^3 R_\xi$)	M_5, k (or M_5^3/k)	One free scale each
Circularity risk	LOW: R_ξ from σ , not from G_N	MEDIUM: k might be fit to G_N	Model A preferred

8 Conclusion

8.1 What v14 Establishes

1. **Model A (compact):** $\mathcal{I} = R_\xi$ with R_ξ from EDC stiffness. Gives $G_N = 1/(8\pi M_5^3 R_\xi)$. Requires calibration of $M_5^3 R_\xi$.
2. **Model B (warped):** $\mathcal{I} = 1/k$ with k the warp rate. Gives $G_N = k/(8\pi M_5^3)$. Requires calibration of M_5^3/k .
3. **EDC content:** Both models use EDC geometry (R_ξ, σ), but neither determines the absolute scale.
4. **Preferred model:** Model A has lower circularity risk because R_ξ is already defined in EDC independently of gravity.

8.2 What Remains Open

Gap	Resolution
Absolute scale (M_5 or $M_5^3 L$)	Calibrate with ℓ_P^{obs} [BL]
Choice between Model A and B	Physics input needed (testable difference?)
M_5 from first principles	Not available in current EDC

8.3 One-Line Outcome

PARTIAL BRIDGE: $\mathcal{I}$ computed up to one EDC parameter (R_ξ or k);
requires 1 calibration scale (ℓ_P or M_{Pl}).
Preferred: Model A with $\mathcal{I} = R_\xi$.

References

1. L. Randall and R. Sundrum, Phys. Rev. Lett. **83**, 3370 (1999).
2. L. Randall and R. Sundrum, Phys. Rev. Lett. **83**, 4690 (1999).
3. R. Maartens, Living Rev. Relativ. **7**, 7 (2004).